

Biological Quasicrystals and the Golden Mean: Relevance to Quantum Computation

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ABSTRACT

Since the discovery of metallic quasicrystals, which lack translational symmetry, much work has been done on their characterisation. In particular mathematical aperiodic tilings have been invoked in an effort to explain their existence. It appears that in three-dimensional quasicrystals non-local quantum effects may not be required. However, here I present some instances - ribosome organisation in embryonic plants, neurotransmitter receptor complexes in animal nervous systems, the cross-section of microtubule bundles and nucleic acids - where biological two-dimensional quasicrystals may occur. Two-dimensional quasicrystals do require non-local quantum effects. This has possible ramifications for both plant development and consciousness. The golden mean is integral to the structure of aperiodic tilings and is found widely in nature, including in plant development, brain EEG waves, and in the consciousness of beauty. I suggest that somehow the encoding of the golden mean in sub-cellular quasicrystals leads to its expression in these macro systems.

Key Words: quasicrystal, golden mean, consciousness, plant development, non-local quantum effects

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Introduction

Despite the many papers that have been written on the function of the nervous system and brain, we still do not have definitive answers to many questions, including how consciousness arises. Similarly we do not fully understand how plant growth and form develops. These two “hard problems” may actually be ultimately insoluble. The fractal nature of these processes might not allow a full understanding (Gardiner, 2013) but if quantum effects are necessary in both instances this may also preclude complete theories. I previously suggested that ribosomal quasicrystals might play a role in plant

“consciousness” (Gardiner, 2012) and here I extend this work, providing further evidence of their existence. I then extrapolate to animal consciousness and show that neurotransmitter receptor complexes may similarly form quasicrystals. As 2-dimensional quasicrystals appear to require non-local quantum effects to exist the ribosomes and neurotransmitter receptors may act as quantum bits, “qubits”, thus allowing quantum computing in both systems. The golden mean is present in both plant development and consciousness (e.g., EEG) and this is thus another correlate of these two processes. I speculate that these macro-effects may be due to the intrinsic presence of the golden mean in quasicrystal mathematics.

Quasicrystals

In 1984 Dan Schechtman discovered quasicrystals (Schechtman *et al.*, 1984), a form of matter that displays an electron diffraction pattern with icosahedral symmetry. This form of

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symmetry is incompatible with atomic periodicity and therefore cannot exist for periodic crystals. Various models of their atomic structure were proposed including aperiodic tiling models based on two and three Penrose tiles and their variants. The Penrose tiling consists of two basic tiles, the kite and the dart (Figure 1).

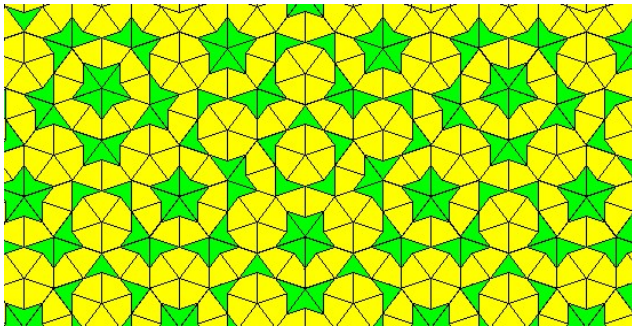


Figure 1. A Penrose tiling showing aperiodicity, with kites in yellow and darts in green. Local pentagonal order is present.

Along the edges of each are bumps and dents which encode local matching rules (Dworkin and Shieh, 1995). A key aspect of such aperiodic tilings is that they do not possess translational symmetry, unlike regular or periodic tilings.

With crystals the generation of long-range order poses no conceptual difficulty since the kinetics can be treated at level of aggregation of unit cells. For quasicrystals the manner in which growth is to be continued may be determined by arbitrarily distant regions of the quasicrystal and it thus appeared that non-local interactions had to be invoked. Penrose (1989) suggested that since there are non-local aspects to quantum theory, these may be being exploited in the way that a quasicrystal grows. However, if growth can be described by addition of geometric units representing clusters of atoms and we do not worry about the formation of the clusters, then an algorithm can describe the formation of quasicrystals using “local” rules only (Socolar, 1999). It must be pointed out that this is an idealised picture of quasicrystal formation and it is not yet known whether the formation of quasicrystals in reality is governed by such rules. The meaning of the term “local” in this study is also not the same as that intended by Penrose (1989).

More recently molecular simulations have shown that the aperiodic growth of three-

dimensional quasicrystals is controlled by the ability of the growing quasicrystal nucleus to incorporate kinetically trapped atoms with minimal rearrangement. In the system studied stable icosahedral clusters are assimilated. For small nuclei and long-range ordering it appears that different growth mechanisms are at work (Keys and Glotzer, 2007). Tsai-type clusters of the quasicrystal approximant CaCd_6 emerge from an atomic-size-driven transformation from planar arrangements to spherical clusters (Berns and Fredrickson, 2013), perhaps reminiscent of the formation of clathrin cages from triskelia. However, these studies of quasicrystal growth are for three-dimensional quasicrystals and the biological quasicrystals of interest in this study are two-dimensional. While it appears that local rules can dictate the structure of three-dimensional quasicrystals, in two-dimensional instances non-local rules are required (Dworkin and Shieh, 1995; van Ophuysen, 1998).

Hydrogen bonding is important in nucleic-acid base pairing and in determining the secondary, tertiary, and quaternary structure of proteins. Recently hydrogen bonding between carboxylic acid functional groups in self-assembled monolayers of ferrocenecarboxylic acid (FcCOOH) has been shown to spontaneously form two-dimensional quasicrystalline arrays. Cyclic hydrogen-bonded pentamers combine with dimers so as to exhibit local five-fold symmetry and maintain translational and rotational order without periodicity for distances of more than 400 angstroms (Wasio *et al.*, 2014). Other processes are involved in protein structure (particularly tertiary and quaternary structure), including sulphur bridges between cysteine side chains, ionic binds and van der Waals dispersion forces. Thus a mechanism present in protein folding has the potential to lead to quasiperiodicity.

Geometry in Biological Systems

Geometry is important in many aspects of protein quaternary structure. The membrane skeleton of human erythrocytes forms a primarily hexagonal lattice of junctional F-actin complexes cross-linked by spectrin filaments (Liu *et al.*, 1987). In tubular crystals of acetylcholine receptor complexes from *Torpedo marmorata* the neurotransmitter receptor complexes form ribbons of paired molecules, a pentagonal tiling, which appears to be due to contact between δ -

subunits (Brisson and Unwin, 1984). Acetylcholine receptors can also take different quaternary structure depending upon the biological system and other interacting proteins (see below). Pentagons can form periodic tilings known as Fibonacci pentilings. These tilings are periodic, but share some characteristics with aperiodic tilings (Caspar and Fontano, 1996). C-reactive protein, one of the classical members of the petraxin family, when crystallised on lipid monolayers by specific adsorption forms a clearly defined pentiling (Wang and Sui, 2001). Clathrin triskelia self-assemble into a limited selection of fullerene cages. Here a geometric constraint – exclusion of head-to-tail dihedral angle discrepancies explains the limited selection of forms and how the cages can form in the first instance (Schein and Sands-Kidner, 2008).

A few biological three-dimensional quasicrystals have been found to exist. The Boerdijk-Coxeter helix is a linear stacking of regular tetrahedra and a quasicrystalline structure with $1+\sqrt{3}$ edges per turn. Protein helices wind around the Boerdijk-Coxeter helix, i.e. the α -helix which is the most common secondary structure of native proteins. Collagen is an example of such a biological quasicrystal (Sadoc and Rivier 2000). Indeed a quasicrystal model of collagen microstructure has been put forward based up second harmonic generation microscopy of the protein (Xu *et al.*, 2010), providing experimental support for collagen being a quasicrystal. Core proteins of human T-lymphotropic leukemia retroviruses show a quasiperiodic primary structure of α -helical and δ -helical segments, the latter being similar to the quasiperiodic proteins of collagen fibres (Liquori *et al.*, 1987). Capsid protein complexes of certain viruses possess regions with a chiral pentagonal order of protein positions in a structure commensurate with dodecahedral geometry (Konevtsova *et al.*, 2012).

Biological Two-Dimensional Quasicrystals

The presence of lipid monolayers and bilayers in biological systems offers unique opportunities for the development of two-dimensional quasicrystals of membrane-associated proteins. Similarly to the pentagonal quasicrystals of ribosomes seen in *Pteridium aquilinum* eggs (Duckett, 1972; Figure 2A) pentagonal rosettes of ribosomes arranged in a hexagonal arrays have been seen in embryonic apple (*Pyrus malus* L.)

leaves cooled for 8 months at 0°C to break dormancy (Bouvire-Durand *et al.*, 1981; Figure 2B).

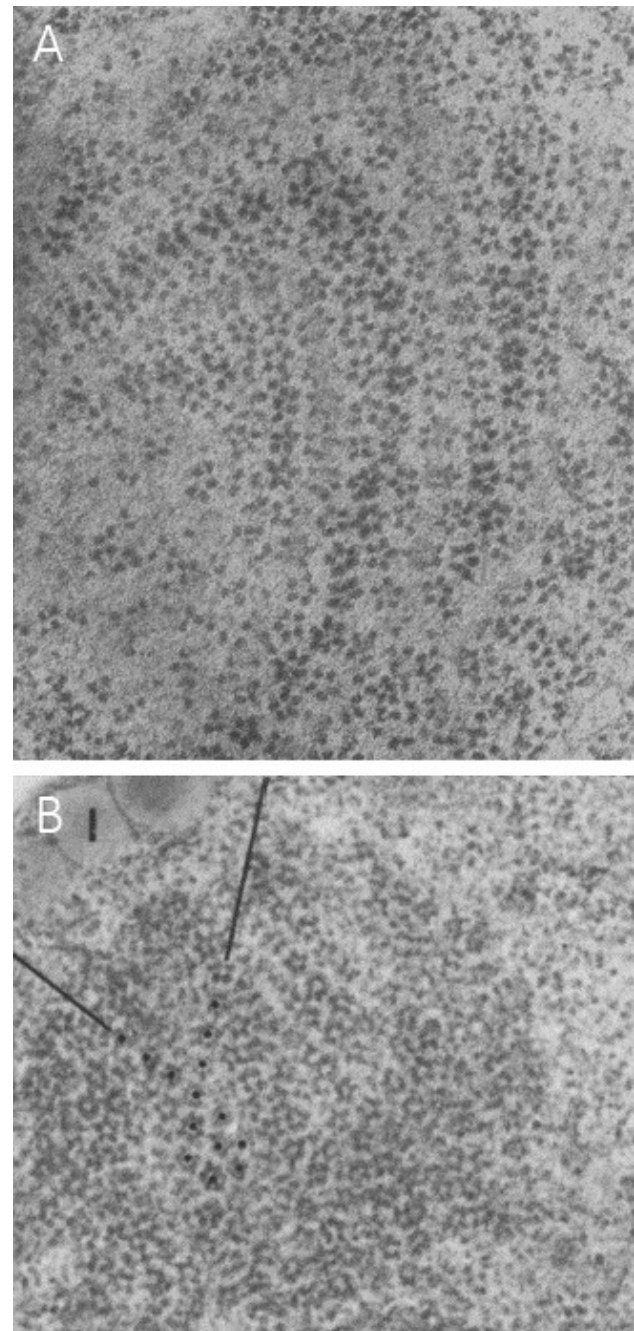


Figure 2A and B. Arrays of pentagonal ribosomes in fern (Duckett 1972; 2A) and apple (Bouvire-Durand *et al.*, 1981; 2B). The arrays have local pentagonal symmetry but tetragonal global symmetry in the fern and hexagonal global symmetry (lines show divergence of ribosome clusters at 60°) in the apple.

This demonstrates that the quasicrystalline arrangement of ribosomes is not confined only to ferns, but is present in “higher” plants as well. In both instances, although there is a crystalline

array with four-fold or six-fold symmetry, the pentagonal shape and arrangement of the ribosome clusters means they do not have translational symmetry.

The endoplasmic reticulum and associated ribosomes is involved in dormancy release with the endoplasmic reticulum stress and unfolded protein response impacting on seed and bud dormancy release in the peach, *Prunus persica*. This was shown by chilling-induced expression of 11 genes involved in these processes (Fu *et al.*, 2014). Ribosomes connect to the rough endoplasmic reticulum through a defined binding site (Bernabeau and Lake 1982) and are important in plant development. Mutation of the cytosolic ribosomal gene *RPS10B* causes defects in the formation and separation of shoot lateral organs (Stirnberg *et al.* 2012). Leaf adaxial identity, and embryo development are both affected by mutation of Arabidopsis ribosomal proteins. When the ribosomal protein RPL27aC is mutated seed shape changes, embryogenesis is delayed, there are defects in inflorescence and floral meristem development, and leaf patterning is affected (for review see Szakonyi and Byrne, 2011). This maybe through the function of ribosomes in protein synthesis, of they may be “moonlighting” in a different role.

A number of animal neurotransmitter receptors form pentameric, indeed pentagonal, protein complexes. These include the GABA_A (gamma-aminobutyric acid) receptor (Miller and Aricescu, 2014) and the serotonin 5-HT₃ (serotonin) receptor (Hassaine *et al.*, 2014). Pentagonal nicotinic acetylcholine receptor complexes are capable of forming differing supra-complex configurations depending upon local conditions. Tubes from the electric organ of the ray *Torpedo marmorata* sometimes display a parallel, linear configuration of receptor complexes (Brisson and Unwin, 1985). Some tubes, and flared regions at the tube ends show a more “random” configuration. Hexagonal and tetragonal arrays of acetylcholine receptors have been seen in chilled *Torpedo californica* electric tissue. The receptor complexes retain their pentagonal dimensions but are randomly oriented within the tetragonal or hexagonal arrays (Giersig *et al.*, 1989).

Hexagonal arrays of nicotinic acetylcholine receptor complexes have more recently been observed at the developing neuromuscular synapse of *Xenopus laevis*. These aggregates are

of about 10 nm spacing and brought about by agrin stimulation (Kunkel *et al.*, 2001). Cells treated with laminin showed a more random array of receptor complexes. The proteoglycan agrin causes the formation and stabilisation of neurotransmitter receptor scaffolds in concert with rapsyn and MuSK, amongst other proteins (Bowe and Fallon 1995; Rimer 2010). The scaffolding protein rapsyn allows neighbouring acetylcholine receptor complexes to connect at angles of either 72°, 180°, or both (Zuber and Unwin, 2013; Figure 3A). The details of rapsyn interactions with the subunits of the acetylcholine receptor complex still remain to be solved but these rules for the association of rapsyn with acetylcholine receptor complexes suggest a quasiperiodic geometry and indeed individual receptor-rapsyn groupings clearly interact in some fashion with their neighbours (Zuber and Unwin, 2013). While not completely conclusive, these lines of evidence suggest that acetylcholine receptor complexes may have an ability to form quasicrystalline arrays *in vivo*, under certain conditions.

The scaffolding protein for glycinergic and GABAergic neurotransmitter receptors, gephyrin, was proposed to form a hexagonal lattice, based upon the trimeric arrangement of the *Escherichia coli* homologue MogA (Liu *et al.* 2000). Gephyrin is proving to be important in neurological disease with rare exonic deletions increasing risk of autism, schizophrenia and seizures (Lionel *et al.*, 2013). It has been suggested that a rapid increase in neuron hyperexcitability (causing a propensity for epilepsy) could be due to a decline in the number of physiologically active GABA_A receptors due to a lack of scaffolding protein (gephyrin) involved in the trafficking and anchoring of the receptor complexes (González, 2013). Gephyrin acts as a scaffolding protein for pentagonal glycine receptors as well.

Since gephyrin forms a hexagonal lattice, with neurotransmitter complexes at the midpoints of each strut, this leads to a quasicrystalline formation (Figure 3B).

The receptor complexes have local five-fold symmetry but form a hexagonal array in a similar manner to that seen in plant ribosome clusters in apple embryos. This means that while they form a crystalline array it doesn't have translational symmetry (Heine *et al.*, 2013). The packing of glycine receptors is denser than that of GABA_A receptors, suggesting that either gephyrin linked

to glycine receptors has a smaller mesh size or that gephyrin linked to GABA_A receptors has gaps in the assembly of the gephyrin lattice (Heine *et al.*, 2013). It appears membrane lipids might also play a role in quantum communication between neurotransmitter receptor complexes.

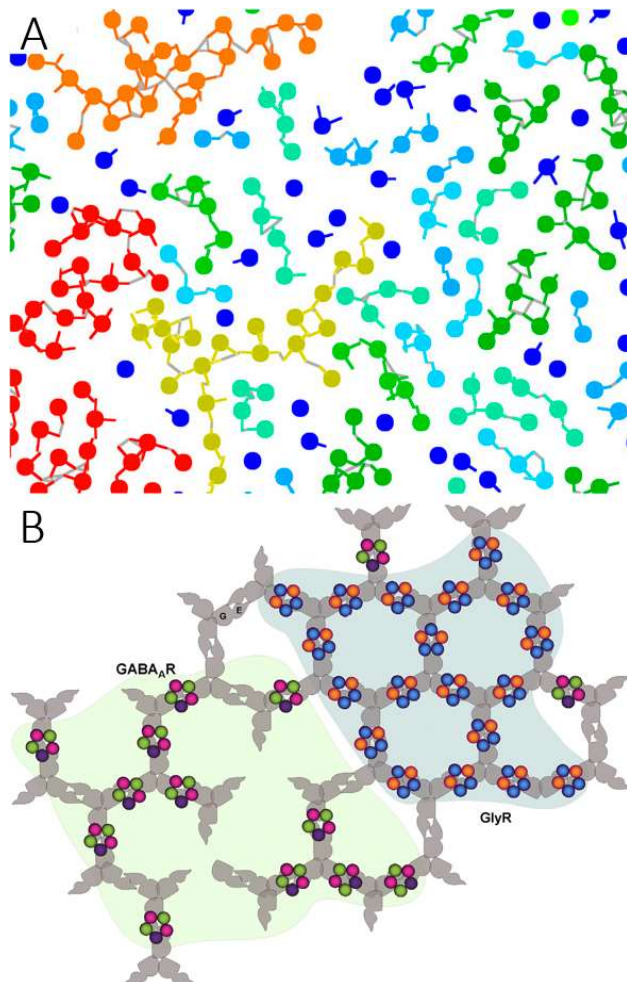


Figure 3A and B. An array of acetylcholine receptors (circles) linked by rapsyn scaffolding proteins (lines). Rapsyn connects to the receptor complexes at points either 72° or 180° apart thus forming a quasicrystalline-like array (Zuber and Unwin, 2013; 3A). Colours indicate number of acetylcholine receptor pentamers in each cluster. Hexameric gephyrin arrays decorated by pentagonal glycine and GABA_A receptor complexes forming, again, a quasicrystalline array with local 5-fold symmetry but overall hexagonal order (Heine *et al.*, 2013; 3B).

Neurotransmitter complexes at the synapse are highly dynamic. This dynamism appears, at least in the case of GABA_A receptor complexes to be due to lateral diffusion of complexes in the plasma membrane, rather than replenishment from intracellular stores (Thomas *et al.*, 2005). Thus receptor complex-scaffolding protein quasicrystals may be constantly rearranging,

with a rearrangement at one point in the quantum entangled lattice instantaneously affecting all other components. Microtubules, as the substrate for gephyrin, may play a role here when they are depolymerised with a drug, thus disrupting clustering of GABA_A receptors, GABAergic currents in hippocampal neurons are affected (Petrini *et al.*, 2003).

Microtubules have been suggested as a key component of the animal consciousness machinery. Certainly apparently quasicrystalline arrays of microtubules have been seen in various biological systems. When the halophyte plants *Mesembryanthemum crystallinum* (Linn.) and *Carpobrotus edulis* (Linn.) are forced to switch to Crassulacean acid metabolism by treatment with NaCl solution they develop microtubules in their chloroplasts. These microtubules have 10 protofilaments and form a hexagonal array in cross-section, with some gaps (Salema and Brandão, 1978). Microtubules in axopodial bundles of radiolarians, including *Raphidiophyrus ambigua*, show hexagonal arrangements (quasicrystalline) with 13 protofilaments and up to 4 linkers per microtubule (Bardele, 1977; Figure 4) and hexagonal bundles of microtubules have also been seen in neuronal microtubules (Needleman *et al.*, 2004).

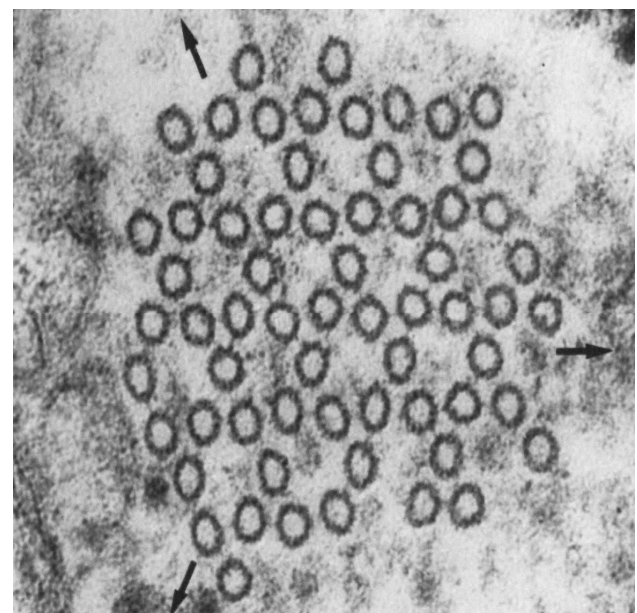


Figure 4. Hexagonal quasicrystalline array of microtubules from *Raphidiophyrus ambigua* (Bardele, 1977). If the observer fixes their eyes to the recurrent X configuration they may envisage swarms of butterflies flying in the direction of the three arrows. Since microtubules are imperfect helices of 13 protofilaments this may lead to the distortions seen in microtubule shape.



Quantum Effects and Biological Two-Dimensional Quasicrystals

Biological systems were initially considered too “warm, wet, and noisy” quantum processes and quantum coherence which previously had been observed only at sub-zero temperatures. However, it has now been shown that quantum coherence exists in photosynthesis at physiological temperatures (Panitchayangkoon *et al.*, 2010). Quantum processes also form the basis for the navigational ability of birds (Cai and Plenio, 2013) and are a key factor in olfaction (Bittner *et al.*, 2012). For a detailed discussion on “the coming of age of quantum biology” see Al-Khalili and McFadden (2014).

All this evidence suggests that quantum effects may in fact be common in biology. Nonlocality is a defining feature of quantum mechanics, but it is beginning to look like the universe is even more non-local than would be expected by quantum mechanical theory (Popescu, 2014). This is perhaps not surprising. Particle entanglement experiments show that particles are non-locally entangled across space-time, showing light is not necessary to convey information non-locally. The big-bang theory says that, at one time, all particles occupied the same space. If true, this means that everything is quantum entangled non-locally (Irwin, 2014). There is similarity here with the thought of the ancient traditions of Buddhism, which also holds that all things are interconnected. Here I propose that quantum effects may be involved in the formation of two dimensional biological quasicrystals, with possible ramifications for plant development and animal consciousness.

Quantum Computing, the Uncertainty Principle and Plant Development

If quantum computing is possible in biological systems, as suggested by Hameroff and Penrose (2003), then it uses qubits (quantum bits) as opposed to binary bits as used by digital computers. Thus it may use quantum-mechanical phenomena such as superposition and entanglement in their operation. Qubits can be in superposition of state. Each qubit can represent a one, a zero, or any quantum superposition of the two states. A quantum computer sets the qubits in an initial state determined by the problem at hand and then manipulates them with a series of quantum logic gates, the sequence of which is called a quantum algorithm. Quantum

algorithms provide the correct solution to the problem only with a certain known probability. This is due to the Uncertainty Principle. In the Copenhagen Interpretation of the Principle, a quantum system is described by a wave function representing the state of the system which smoothly evolves until a measurement is made when it collapses to an eigenstate that is observable. Good progress is being made in developing artificial quantum computers using these principles (Barz *et al.*, 2014).

It has long been believed that plant development is deterministic (Green, 1999). The use of fractal L-system (Lindenmayer system) mathematics whereby formal grammar is used to replicate plant-like structures appeared to offer some hope for a reductionist model of plant development (Prusinkiewicz and Lindenmayer, 1996). However, despite thousands of publications on the subject we still do not understand how this might take place in terms of subcellular organisation. As with the “hard problem” of consciousness quantum non-locality might play a role. A totipotent plant cell embryonic cell exists in a state of potentialities with regard to development. As a waveform collapses another “tile” is added to the quasicrystal ribosome mosaics mentioned earlier. The non-locality of the quasicrystalline ribosome array enables ribosome function to be integrated across its entirety. This then leads to changes in protein translation (and/or other structural changes in the cell) and development subsequently becomes deterministic. The sequence of wave form collapses continues as long as the embryo is determining cell fate. The final outcome of plant growth and development cannot be predicted a priori, it is probabilistic due to non-local quantum effects.

For example, the spiral growth of plants can be either left- or right-handed. This spiral growth appears to have a large non-genetic component (Wu *et al.* 2008):

“All these uncertainties suggest uncertainties in the operation of position-determining mechanisms such as one would expect of biological rather than purely mechanical systems ... One of the consequences of the frequent inequality of even the opposite primordia in a decussate system is the tendency to the development of spirality which may be left-handed or right-handed ... In certain populations ... a preponderance of left-handed coconut palms (*Cocos nucifera*); surprisingly, the left-handed palms were

more vigorous and yielded a larger crop than right-handed palms." (Carr, 1984).

Thus, from an evolutionary point of view, it makes no sense for palms to be right-handed since their fertility is less than that of the left-handed palms. So why do right-handed palms exist? Perhaps because the non-locality required for palm development in the first instance means that there will always be the possibility of a right-handed alternative to the left-handed palm.

There are many examples of non-determinism in plant development: four-leafed clover, flowers with unusual development, trees with unusual spiral wood grain and at the cellular level sometimes there are two pre-prophase bands of microtubules instead of one (Granger and Cyr, 2001). Indeed the TON and FASS mutant Arabidopsis lines show normal cell division and cytokinesis without the presence of a pre-prophase band (Traas *et al.*, 1995), showing the resilience of plant developmental processes to major disruption. This probabilistic nature of development suggests that the Uncertainty Principle and quantum computing may be present. It has even been suggested that the Orch OR hypothesis of human consciousness may be applicable to plants, with large numbers of microtubules present in connecting plant ray parenchyma cells (Peter Barlow, personal communication).

Quantum Computing, the Uncertainty Principle and Consciousness

It is suggested that the brain can operate as both quantum computer and conventional neurocomputer (Hameroff, 2007) and that the quantum computer within the brain invokes non-computable influences from information embedded in space-time geometry, potentially avoiding algorithmic determinism and "rescuing" free will (Hameroff 2012). The Penrose-Hameroff "Orchestrated Objective Reduction" (Orch OR) model suggests that when a sufficient mass of tubulin subunits are quantum-mechanically linked, the quantum waveform will spontaneously collapse, resulting in a conscious moment (Hameroff and Penrose, 2003).

There are some plusses to this theory. Recent work on anaesthesia, which selectively erases consciousness while sparing non-conscious brain activities, acts via a destabilisation of MTs in brain neurons (Emerson *et al.*, 2013). Microtubules contain a unique

arrangement of water molecules in their lumen. These water molecules allow entire brain microtubules to act as a single molecule with the microtubule's thermal and optical properties programmed within a single tubulin molecule (Sahu *et al.*, 2013). Electric pulses can course along microtubules, allowing ultra-fast electric signalling (Havelka *et al.*, 2014). Memory may be stored in microtubule lattices and play a role in quantum computing processes. This could be through the activity of Ca^{2+} /calmodulin-dependent protein kinase (CaMKII) which is a dodecameric holoenzyme containing two hexagonal sets of kinase domains. The hexagonal sets of kinase domains seem possibly positioned for interaction with the hexagonal array of tubulin subunits within a microtubule. Thus phosphorylation of tubulin subunits offers the potential for vast storage of information (Craddock *et al.*, 2012). Other posttranslational modification of microtubules may encode information which can then be "read" by microtubule-associated proteins (Verhey and Gaertig, 2007). Indeed microtubules are intrinsic to biological memory, which is possibly a fundamental of consciousness (Gardiner, 2012). It looks possible that quasicrystalline neurotransmitter receptors and microtubules are intimately connected with regard to electrical signalling in the nervous system, and thus consciousness (Gardiner *et al.*, 2011).

There are some potential difficulties with the Penrose-Hameroff Orch OR hypothesis too. Orchestrated quantum reduction was originally posited to avoid the "unsettling" implications of quantum-coherent superpositions persisting until conscious observation collapses the waveform, or "subjective reduction". But is this just because quantum mechanics is often counterintuitive and we perhaps still do not fully understand it? To counter this "unsettlement" various schemes for objective reduction have been proposed see Hameroff and Penrose (2003). There are, however, some major objections to orchestrated reduction. For example, such theories postulate that determinate measurement outcomes are grounded in particular structures within the wave-function. But the wave-function has many such structures so a symmetry breaker is required to distinguish between favoured structures from unflavoured structures. The only relevant differences between them are mod-square values and there is no reason why such values should be symmetry



breakers (McQueen, 2015). The necessity for water to be present in microtubules in order to allow a microtubule of many tubulin dimers to act as a single molecule also suggests that modifications need to be made to this theory.

Can two-dimensional biological quasicrystals act as quantum computers? It seems likely that non-local quantum effects are necessary for their existence and the “decision” of where to place the next tile in the formation requires such effects. The quantum bits, or “qubits” are encoded by the pentagonal ribosomes or neurotransmitter receptor clusters respectively. This suggests that protein complexes, whether ribosomes or neurotransmitter receptors or microtubules, are in instantaneous communication across considerable subcellular distances. In the next sections I examine whether the presence of the golden mean in biological systems supports this hypothesis.

The Beauty of Symmetry and the Golden Mean in Biology

The golden mean and Fibonacci sequence crop up time and time again in the mathematics of two-dimensional quasicrystals. In a correct Penrose tiling “worms” are straight lines in the tiling, in the neighbourhood of which the differences between the large-scale dart and kite will first appear. “Worms” are made up of two kinds of units, the long and the short. The sequence of long and short, for a correct tiling, always follow a rule given by the Fibonacci sequence. The presence of the Fibonacci sequence here in a correct tiling means that the tiling must be highly non-local (Penrose, 1989). The golden mean is also the ratio of chord lengths to side lengths in a regular pentagon. This leads to the observation that the ratio of long side lengths to short in both kite and dart tiles of a Penrose tiling is also the golden mean. In pentilings, as the number of pentagons in each translational region increases and the pentiling approaches a true quasicrystalline form, the packing density of the pentagonal subunits approaches the golden mean/2 (Caspar and Fontano, 1996).

Symmetry is seen as beautiful and interesting by humans, as evidenced by its presence in mathematics dating back to at least 400 BC. Indeed the Greek word *symmetros* was originally meant to convey the meaning “well proportioned” or “harmonious” (Bollinger *et al.*

2001). Often architecture and art have been in the vanguard of human mathematics. Gothic cathedrals have been suggested to reflect the underlying fractal nature of human consciousness due to their porous, scale-free structures. Indeed the first known example of the fractal known as the Sierpinski triangle is found in a floor mosaic of the cathedral in Anagni, Italy, dating to 1104 AD – well before its appearance in the discipline of mathematics (Goldberger, 1996). Islamic architecture appears to have discovered quasiperiodicity as early as 1197 on the external walls of the Gunbad-i Kabud tomb tower in Maragha, Iran, and 1323 in the construction of the Madrasa of al-‘Attarin in Fez, Morocco (Al Ajlouni, 2012). The existence of these geometric principles in early art and architecture might suggest they play a role in our perception of what is “well proportioned” and “harmonious”.

The golden mean (phi, named after Phidias who has been suggested to be the first to use it in art) has long been acknowledged as somehow appearing beautiful to the human consciousness. Euclid was the first to define it mathematically, by dividing a line AB at a point C such that CB is to AC as AC is to AB. The golden mean is irrational and has the value of 1.6180339... The Fibonacci numbers are simply the arithmetic consequence of multiplying each number by the golden mean and rounding to the nearest integer. The Fibonacci sequence also has the property that each number in the sequence is the sum of the two previous numbers, thus 1,1,2,3,5,8,13,21, etc. With the rediscovery of ancient Greek and Roman knowledge in the Renaissance, many artists including Poussin and Piero della Francesca began again to use it in their work. Modern day artists including Seurat, Le Corbusier, Mondrian and the Australian Jeffrey Smart (an admirer of della Francesca), have continued this tradition (Rigby, 2002; Figure 5).

The golden mean is present even in the oscillations of brain waves. The classical frequency bands of the EEG can be described as a geometric series with a ratio between neighbouring bands of 1.618. This means that the frequencies of brain waves will never synchronise (Pletzer *et al.*, 2010). The presence of the golden mean here provides the most desynchronised possible state of brain waves, the possibility of spontaneous and most irregular coupling between waves and the possibility for transition from a resting state to activity (Pletzer *et al.*, 2010). The golden mean appears in



pharmacology. The optimum concentration for determining whether two channel blockers act at the same site (Syntopic Model) or different sites (Allotopic Model) is that, when used alone, they have an inhibitory effect of close to 61.8%, which is the reciprocal of the golden mean. This has been experimentally shown with 5-HT₃ receptors and the golden mean also appears in the simplest of pharmacological equations, the Hill-Langmuir equation (Jarvis and Thompson, 2013).



Figure 5. "Classical Landscape" by John Glover c. 1820, Art Gallery of New South Wales Collection. The ratio of the canvas to its width is 0.61 (reciprocal of the golden mean). The clump of trees to the left is centred at about 0.61 of the width of the canvas and the horizon is at about 0.63 of the canvas depth from the top.

The golden ratio is seen during many stages of plant development. It is well known to occur in spiral development of flowers and branching of trees. Two recent papers have extended the paradigm, with cell shape proportions of *Arabidopsis* pavement cells defined by length:width ratios measured over 2,400 cells being close to the golden ratio (Staff *et al.*, 2012). Stomata here also appear to be positioned by a golden ratio-dependent means. The form of the *Arabidopsis* seed too appears to be dependent upon the golden mean. Comparison of the outline of its longitudinal section with a transformed cardioid, transformed by scaling the horizontal axis by a factor equal to the golden mean, approximates the shape of the seed more accurately than other figures (Cervantes *et al.*, 2010).

The golden mean is important in the secondary structure of RNA and DNA sequences. Statistical analysis of RNA secondary structures of all 480 sequences from RNA STRAND, which are validated by NMR or X-ray, finds that the length ratios of domains in these sequences are

approximately 0.382L, 0.5L, 0.618L (inverse of golden mean), and L, where L is the sequence length. This has enabled the development of an algorithm capable of detecting RNA pseudoknots which is more accurate than previous algorithms (Li *et al.*, 2014). Two attractors in the 6 folding steps of DNA codon populations show the evidence of two attractors. The numerical relationship between the attractors is derived from the golden mean (Perez, 2010). These findings are of particular interest as the information for retrieving the normal reading frame of DNA implies the presence of short-range correlations and almost periodic structures, offering an analogy with the properties of quasicrystals (Giannerini *et al.*, 2012).

The Golden Mean and Quasicrystals

I have shown that the golden mean is present in processes based upon subcellular two-dimensional quasicrystals – both plant development and brain function, particularly EEG brain-waves and the perception of beauty by the conscious mind. I believe this may be due to the expression of subcellular information in the macro. Thus, I suggest, the presence of biological quasicrystals at the subcellular level may somehow give rise to the expression of the golden mean in consciousness, plant development, and elsewhere. Certainly plant ribosomes, which are a two-dimensional quasicrystalline array in the embryo, play a key role in plant development through protein translation. A recent *in vitro* study has demonstrated how geometry can be very important in the process of protein translation (Karzbrun *et al.*, 2014). Similarly, neurotransmitter receptor complex quasicrystals play a role in brain waves (where the golden mean is again seen) and in our consciousness which somehow sees the golden mean as aesthetically attractive. Indeed the perception of beauty leads to greater activation of EEG waves in the brain (Lengger *et al.*, 2007). Even the structure and function of nuclei acids may be governed by quasicrystal-like properties and the golden mean. And then there are the ubiquitous microtubules, important in both plant development and consciousness again possibly forming quasicrystalline arrays. This may allow quantum, non-local, coordinated electrical conduction along bundles of microtubules in neurons and elsewhere.

Conclusions

I realise that the above hypothesis is highly speculative. But to date there has been little in the way of hypotheses as to how the golden mean crops up in biological systems. It is clearly important, and its presence in both plants and animals would tend to suggest a commonality. The Orc OR hypothesis of consciousness is, I believe, probably the best currently available. I

do not believe it is fully correct but this is how theoretical biology works. Each successive theory takes us closer to the truth, although the ultimate truth may be in the end elusive. I hope that this piece of work may play a similar role, and somewhat increase our knowledge of what it means to be conscious.

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